Supplementary material

Table 1: Basic characteristics of the included case reports and case series

Study	Study type	Ethnicity	Patients included	Age	Female patients	NIHSS	aPTT (sec)	Dabigatran dosage	Stroke to idarucizumab (min)	Alteplase dosage (mg/Kg)	EVT	Hemorrhagic transformation	Death	SICH	SICH definition	INR	Hemo	TT
Fang, C. W., et al. (2019)	Case series	Asian	10	71,1	4	16	30,6				1	3	1	1	ECASS 3			
Frol et al, 2020	Case series	Caucasian	19	75	9	9	40,35	7pt:150mg 12pt:110mg	128	0,9	0	4	2	1	ECASS 2			
Kermer, P., et al. (2017)	Case series	Caucasian	19	7	11	7	42,3	5pt: 150mg, 14pt: 110mg		0,9		0	1	0				
Kermer, P., et al. (2020).	Case series	Caucasian	80	75,9	29	9,7	42,7	32pt:150mg, 48pt: 110mg		0,9		0	3	0				104,5 (63,9)
Barber, P. A., et al. (2020).	Case series	Caucasian	51	73,3	14	8					8	2	3					
Beharry, J., et al. (2020).	Case series	Caucasian	13	79	4	6					7	2	2	0				80 (57-113)
Šaňák, D., et al. (2018).	Case series	Caucasian	13	70	5	7	38,1	8pt: 150, 5pt: 110 mg	427	0,9	0	2	3	1	SITS-MOST			72,3 ±56,1
Tse, D. M., et al. (2018)	Case series	Caucasian	6	68	3	21		4 pt 150 mg, 2 pt 110 mg	180	0,9	2	1	1	1	IST-3 modified			
Küpper, C., et al. (2019).	Case series	Caucasian	7							0,9	4	0	2	0				
Agosti, S., et al. (2017)	Case report	Caucasian	1	71	1	9	29,0	150	210	0,9	0	0	0	0		1,31		
Alvarez Bravo, G., et al. (2017).	Case report	Caucasian	1	65	1	19	31,0	150			0	0	0	0				
Baule, A., et al. (2018).	Case report	Caucasian	1	89	1	4	38,8	110				0	0	0				
Berrouschot, J., et al. (2016).	Case report	Caucasian	1	76	0	11	73,3	110	150		0	0	0	0				
Binet, Q., et al. (2018)	Case report	Caucasian	1	55	1	20	37,4	150	120		1	0	0	0		1,18		
Bissig, D., et al. (2017).	Case report	Caucasian	1	69	0	12	39,2	150	205	0,9	0	0	0	0		1,12		

Candelaresi, P., et al. (2020).	Case report	Caucasian	1	73	1	21		150	115	0,9	1	0	0	0				
Gawehn., et al. (2016)	Case report	Caucasian	1	75	0	5	39,0	110	80	0,9	0	0	0	0	1,0			
Harsha, K. J., et al. (2019)	Case report	Caucasian	1	74	1	11		150		0,9	0	0	0	0				
Hieber, M. and J. Bardutzky (2019).	Case report	Caucasian	1	81	0	6		150	540	0,9	0	0	0	0				
Hosoki, S., et al. (2018)	Case report	Asian	1	74	0	6	41,0	110		0,6	0	0	0	0				
Jala, S. and E. O'Brien (2019).	Case report	Caucasian	1	77	0	11	51,8	150	93	0,9	0	0	0	0	1,5			
Kafke, W. and P. Kraft (2016).	Case report	Caucasian	1	75	1	7	35,5	110		0,9	0	0	0	0	1,1	90	>150	
Lin, Y. T., et al. (2020)	Case report		1	71	0	9		150	189	0,9	1				1,07			
Lo, W. T., et al. (2018).	Case report	Asian	1	78	1	34	50,7	110	115	0,6	0	0	0	0			101	
Loh, C. H. and G. Herkes (2019).	Case report	Caucasian	1	77	0	8	28,6	110	183	0,6	0	0	0	0	1,1	<35	14	
Maramattom, B. and J. Thomas (2019	Case report	Caucasian	1	38	0			150	260	0,9	0	0	0	0				
Meyer, D., et al. (2019).	Case report	Caucasian	1	73	0	19	27,0		35	0,9	0	0	0	0	1,1		12,5	
Mutzenbach, J. S., et al. (2016).	Case report	Caucasian	1	68	0	3	34,0	110	100	0,9		0	0	0	<1,2	34,1		
Ohtani, T., et al. (2019).	Case report	Asian	1	67	1	7	68,0	110	264	0,9	0	0	0	0				
Ohya, Y., et al. (2018).	Case report	Caucasian	1	57	0	22	41,3	110	78	0,9	0	0	0	0				
Renard, A., et al. (2018).	Case report	Caucasian	1	74	0	16		110		0,9	1	1	0	0				
Schäfer, N., et al. (2016).	Case report	Caucasian	1	67	1	10		150	85	0,9		0	0	0			130	
Schulz, J. G. and B. Kreps (2016)	Case report	Caucasian	1	76	0	11	72,2	110		0,9		0	0	0				
Ting, A., et al. (2019).	Case report	Caucasian	1	68	0	7	47,0	150		0,9	0	0	0	0	1,3	193	16	
Tireli, D., et al. (2017).	Case report	Caucasian	1	71	0	6	62,0	150	109	0,9		0	0	0	1,3			

Tsai, L. K., et al. (2018).	Case report	Asian	1	78	1	24	26,8	110	118	0,9	0	1	1	1			
Tsai, Y. T., et al. (2018).	Case report	Asian	1	57	0	9	25,1	110	62	0,9	0	1	0	0			
Turine, G., et al. (2017).	Case report	Caucasian	1	85	1	17	32,2	110	298	0,9	0	0	0	0	1,32		112
von Wowern, F., et al. (2017).	Case report	Caucasian	1	78	1	11	49,0	150			0	0	0	0			
Vukorepa, G., et al. (2020)	Case report	Caucasian	1	80	1	8		110	258	0,9	0	0	0	0			
Ng et al, (2017)	Case report	Caucasian	2	46	0	5				0,9	0	0	0	0			
Pikija et al, (2017)	Case report	Caucasian	5	88	1	10	71,0	110		0,9		0	0	0		202,4	
				67	0	4	77,0	150		0,9		0	0	0		183,7	
				84	0	10	84,0	150		0,9		0	0	0		31,4	
				85	0	7	36,0	110		0,9		0	0	0		43	
				82	0	18	52,0	110		0,9		0	0	0		172,2	
Vosko et al, (2017)	Case report	Caucasian	3	68	0		34,0	110	110	0,9		0	0	0		34	
				84	0	9		110		0,9		0	0	0		79	129
				78	0		134,0	110		0,6		0	0	0			
Gianndrera et al, (2019)	Case report	Caucasian	2	78	0	12	47,0	110		0,9	0	0	0	0			
				85	0	22	53,0	110		0,9	1	1	1	1			
Zhao et al, (2019)	Case report	Caucasian	3	71	0	3	20,0		101	0,9	0	0	0	0	1		12,8
				85	1	35	31,0		134	0,9	1	0	0	0	1		12
				84	1	14			258	tenectepla se	0	0	0	0			
Hieber et al, (2018)	Case report	Caucasian	3	77	0	5		110	45	0,9	0	0	0	0			96,7
				48	0	18		110	75	0,9	1	0	0	0			42,3
				56	0	7		150		0,9	0	0	0	0			53,6
Cappelari et al. (2017)	Case report	Caucasian	1	75	1	4		150				0	0	0			
Facchinetti et al, (2017)	Case report	Caucasian	1	78	1	4		150				0	0	0			

Table 2: Quality evaluation

Study	Selection	Ascertainment		Causality				Reporting	Overall quality
	Does the patient(s) represent(s) the whole experience of the investigator (centre) or is the selection method unclear to the extent that other patients with similar presentation may not have been reported?	Was the exposure adequately ascertained?	Was the outcome adequately ascertained?	Were other alternative causes that may explain the observation ruled out?	Was there a challenge/rechallenge phenomenon?	Was there a dose–response effect?	Was follow-up long enough for outcomes to occur?	Is the case(s) described with sufficient details to allow other investigators to replicate the research or to allow practitioners make inferences related to their own practice?	
Fang, C. W., et al. (2019)	good	good	good	yes	no	no	yes	yes	good
Frol et al, (2020)	good	good	good	yes	no	no	yes	yes	good
Kermer, P., et al. (2017)	good	good	good	yes	no	no	yes	yes	good
Kermer, P., et al. (2020).	good	good	good	yes	no	no	yes	yes	good
Barber, P. A., et al. (2020).	good	good	good	yes	no	no	yes	yes	good
Beharry, J., et al. (2020).	good	good	good	yes	no	no	yes	yes	good
Šaňák, D., et al. (2018).	good	good	good	yes	no	no	yes	yes	good
Tse, D. M., et al. (2018)	good	good	good	yes	no	no	yes	yes	good
Küpper, C., et al. (2019).	good	good	fair	yes	no	no	yes	yes	fair
Agosti, S., et al. (2017)	good	good	fair	yes	no	no	yes	yes	fair

Alvarez Bravo, G., et al. (2017).	good	good	good	yes	no	no	yes	yes	good
Baule, A., et al. (2018).	good	good	good	yes	no	no	yes	yes	good
Berrouschot, J., et al. (2016).	good	good	good	yes	no	no	yes	yes	good
Binet, Q., et al. (2018)	good	good	good	yes	no	no	yes	yes	good
Bissig, D., et al. (2017).	good	good	good	yes	no	no	yes	yes	good
Candelaresi, P., et al. (2020).	good	good	good	yes	no	no	yes	yes	good
Gawehn., et al. (2016)	good	good	good	yes	no	no	yes	yes	good
Harsha, K. J., et al. (2019)	good	good	good	yes	no	no	yes	yes	good
Hieber, M. and J. Bardutzky (2019).	good	good	good	yes	no	no	yes	yes	good
Hosoki, S., et al. (2018)	good	good	good	yes	no	no	yes	yes	good
Jala, S. and E. O'Brien (2019).	good	good	good	yes	no	no	yes	yes	good
Kafke, W. and P. Kraft (2016).	good	good	good	yes	no	no	yes	yes	good
Lin, Y. T., et al. (2020)	good	good	good	yes	no	no	yes	yes	good
Lo, W. T., et al. (2018).	good	good	good	yes	no	no	yes	yes	good
Loh, C. H. and G. Herkes (2019).	good	good	good	yes	no	no	yes	yes	good
Maramattom, B. and J. Thomas (2019	good	good	fair	yes	no	no	yes	yes	fair
Meyer, D., et al. (2019).	good	good	good	yes	no	no	good	yes	good
Mutzenbach, J. S., et al. (2016).	good	good	good	yes	no	no	good	yes	good
Ohtani, T., et al. (2019).	good	good	good	yes	no	no	good	yes	good
Ohya, Y., et al. (2018).	good	good	good	yes	no	no	good	yes	good

Renard, A., et al. (2018).	good	good	good	yes	no	no	good	yes	good
Schäfer, N., et al. (2016).	good	good	fair	yes	no	no	good	yes	fair
Schulz, J. G. and B. Kreps (2016)	good	good	good	yes	no	no	good	yes	good
Ting, A., et al. (2019).	good	good	good	yes	no	no	good	yes	good
Tireli, D., et al. (2017).	good	good	good	yes	no	no	good	yes	good
Tsai, L. K., et al. (2018).	good	good	fair	yes	no	no	fair	yes	fair
Tsai, Y. T., et al. (2018).	good	good	good	yes	no	no	good	yes	good
Turine, G., et al. (2017).	good	good	good	yes	no	no	good	yes	good
von Wowern, F., et al. (2017).	good	good	good	yes	no	no	good	yes	good
Vukorepa, G., et al. (2020)	good	good	good	yes	no	no	good	yes	good
Ng et al, (2017)	good	good	good	yes	no	no	good	yes	good
Pikija et al, (2017)	good	good	good	yes	no	no	good	yes	good
Vosko et al, (2017)	good	good	fair	yes	no	no	yes	yes	good
Gianndrera et al, (2019)	good	good	good	yes	no	no	yes	yes	good
Zhao et al, (2019)	good	good	good	yes	no	no	yes	yes	good
Hieber et al, (2018)	good	good	good	yes	no	no	yes	yes	good
Cappelari et al. (2017)	good	good	good	yes	no	no	yes	yes	good
Facchinetti et al, (2017)	good	good	good	yes	no	no	yes	yes	good
Laxamana et al (2020)	good	good	good	yes	no	no	yes	yes	good